

Meteorites, their masses and geolocation:
Analysis of a NASA dataset

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Introduction

Meteorites are extraterrestrial objects, usually rocks or metals that survived their entry into the earth atmosphere. For this project, I decided to analyse a DataSet from the famous space agency NASA containing all the recorded meteorites found or seen falling on earth.

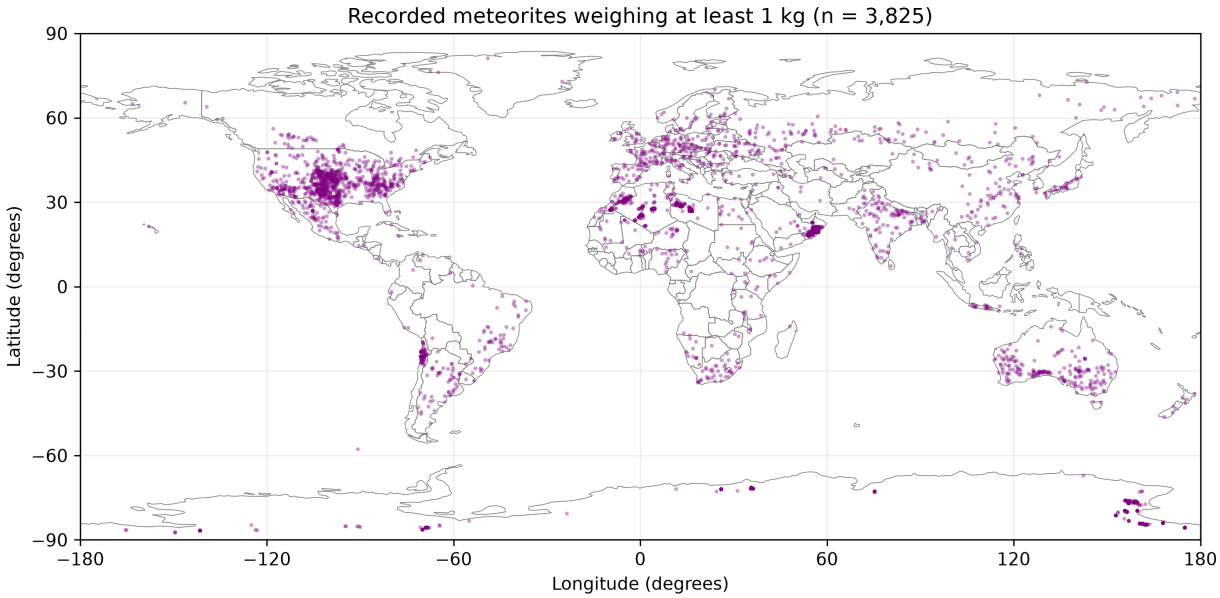
My research question was as follows : **Where are meteorites of masses of at least 1 kg located, how do counts change as the minimum mass increases, and in which countries were found the ten heaviest recorded meteorites ?**

Data sample

The dataset analysed contained 45,715 recorded meteorites described using the following parameters: name,id (number), nametype, recclass, mass (g),fall ("fell" meaning seen falling or "found"),year, reclat (latitude), reclong (longitude) and GeoLocation. For my research question, I was especially interested in the recorded mass, longitude and latitude data. After cleaning for missing mass, location data and meteorites with masses below 1 kg, I was left with only 3,825 meteorites.

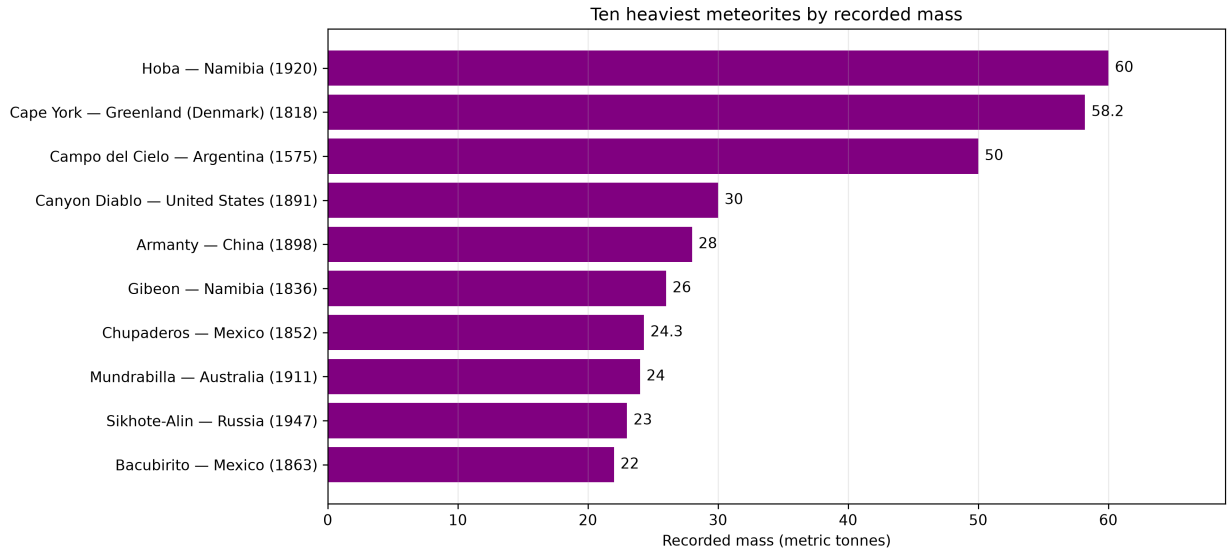
Results and discussion

To begin with, we looked the disposition of the heavy meteorites (more than 1 kg) on a world map to understand if any region was more affected by meteorites fall. As seen from the map, the united states were the winner for meteorites count (928 in total). A possible explanation is that the data analysed were gathered by an American company.



Then, we looked at the count of meteorties for different mass thresholds : 1 kg, 10 kg, 100 kg and one ton. We will display the numbers here to avoid too many graphs, meteorites bigger or equal than : 1 kg = 4,871, 10 kg = 1,388, 100 kg = 305, 1,000 kg = 54. As we can see only 54 out of the 4,871 meteorites heavier than 1 kg weight more than 1 ton which is pretty low and reassuring when thinking about the possible damages such a meteorite could cause when falling.

The next step in our analysis was to come up with a top 10 of the heaviest meteorites recorded, the country where they were found as well as the year. The analysis gave this plot : The heaviest meteorite



recorded was significant in mass, 60 tons found in Namibia and was given the name Hoba for the nearby Farm. It's lenght was around 2,7 m. Next, Cape York found in Greenland Denmark in 1818 and weighting 58.2 tons. The most recent meteorite present in our top ten was found in 1947 in Russia and was given the name Sikhote-Alin.

Conslusion

This analysis was meant to understand meteorites better, especially their mass division and the location of the heaviest ones on earth. The first plot shows a geographically uneven distribution and the count for different thresholds shows a declining number of meteorites for increasing weight. Finally, the last figure shows a top ten of the heaviest meteorites, their name and the country where they were found.

References

[1] NASA Open Data Portal. *Meteorite Landings*. Available at: https://catalog.data.gov/dataset/meteorite-landings.